

**EXPERIMENT NO. : 1 (ASSIGNMENT)**DATE: 13/08/2026

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**1. AIM**

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To design and study a lithography mask pattern for shallow trench isolation, illustrating line space structures within a defined boundary for device isolation for given masks.

**2. TOOL**

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Semulator3D

**3. THEORY**

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Shallow Trench Isolation (STI) is a semiconductor fabrication technique used to electrically isolate adjacent devices. In this process, shallow trenches are formed in the silicon substrate and filled with an insulating material. Lithography is used to define the regions where the trenches are to be formed.

In this assignment, two lithography mask designs are studied. Design 1 consists of a  $500\text{ nm} \times 400\text{ nm}$  boundary containing four vertical mask openings, each  $40\text{ nm}$  wide and separated by  $60\text{ nm}$  spacing. The openings form a regular line-space grating oriented along the Y-axis. Design 2 consists of a  $500\text{ nm} \times 500\text{ nm}$  boundary with a  $60\text{ nm}$  wide horizontal bar and a  $60\text{ nm}$  wide vertical bar, both extending across the complete boundary and intersecting at the center.

These patterns illustrate how lithographic dimensions such as opening width, spacing, orientation, and alignment are used to define regions for device isolation. Precise control of these parameters is important for transferring the required pattern accurately during semiconductor fabrication.

4. DESIGN PARAMETERS

Table 1: assignment 1

| Number | Lock | Action         | Name                   | Material(s)        | Dopant(s) | Depth | Mask | Mask Polarity | Wafer   | Wafer Side | Lateral Ratio | Parameter Name | Comment |
|--------|------|----------------|------------------------|--------------------|-----------|-------|------|---------------|---------|------------|---------------|----------------|---------|
| 1      |      | Wafer Setup    | Wafer Setup            | Silicon/Silicon... |           | 200   |      |               | WAFER_1 |            |               |                |         |
| 2      |      | Conformal...   | pad oxide deposition   | /Oxides/Silicon... |           | 4     |      |               | WAFER_1 | Both       | 1.0           |                |         |
| 3      |      | Conformal...   | pad nitride deposition | /Nitrides/SiN      |           | 40    |      |               | WAFER_1 | Top        | 1.0           |                |         |
| 4      |      | Sequence       | lithography            |                    |           |       |      |               | WAFER_1 |            |               |                |         |
| 4.1    |      | Planarizing... | spin on resist         | /Organics/photo... |           | 120   |      |               | WAFER_1 | Top        |               |                |         |
| 4.2    |      | Expose resist  | Expose resist          | /Organics/photo... |           | 0     | tx   | Light         | WAFER_1 | Top        |               |                |         |
| 5      |      | Sequence       | etch fins              |                    |           |       |      |               | WAFER_1 |            |               |                |         |
| 5.1    |      | Enhanced B...  | pad nitride etch       | /Nitrides/Org...   |           | 40    |      |               | WAFER_1 | Top        | 0.05          |                |         |
| 5.2    |      | Enhanced B...  | pad oxide etch         | /Nitrides/Org...   |           | 4     |      |               | WAFER_1 | Top        | 0.05          |                |         |
| 5.3    |      | Enhanced B...  | silicon etch           | /Nitrides/Org...   |           | 100   |      |               | WAFER_1 | Top        | 0.05          |                |         |
| 6      |      | Remove M...    | resist strip           | /Organics/         |           |       |      |               | WAFER_1 |            |               |                |         |
| 7      |      | Conformal...   | sti fill deposition    | /Oxides/SiO2_H...  |           | 200   |      |               | WAFER_1 | Top        | 0.67          |                |         |
| 8      |      | Polish         | sti oxide cmp          |                    |           |       |      |               | WAFER_1 | Top        |               |                |         |

Table 2: assignment 2

| Number | Lock | Action         | Name                   | Material(s)        | Dopant(s) | Depth | Mask       | Mask Polarity | Wafer   | Wafer Side | Lateral Ratio | Parameter Name | Comment |
|--------|------|----------------|------------------------|--------------------|-----------|-------|------------|---------------|---------|------------|---------------|----------------|---------|
| 1      |      | Wafer Setup    | Wafer Setup            | Silicon/Silicon... |           | 200   |            |               | WAFER_1 |            |               |                |         |
| 2      |      | Conformal...   | pad oxide deposition   | /Oxides/Silicon... |           | 4     |            |               | WAFER_1 | Both       | 1.0           |                |         |
| 3      |      | Conformal...   | pad nitride deposition | /Nitrides/SiN      |           | 40    |            |               | WAFER_1 | Top        | 1.0           |                |         |
| 4      |      | Sequence       | lithography            |                    |           |       |            |               | WAFER_1 |            |               |                |         |
| 4.1    |      | Planarizing... | spin on resist         | /Organics/photo... |           | 120   |            |               | WAFER_1 | Top        |               |                |         |
| 4.2    |      | Expose resist  | Expose resist          | /Organics/photo... |           | 0     | cross_line | Light         | WAFER_1 | Top        |               |                |         |
| 5      |      | Sequence       | etch fins              |                    |           |       |            |               | WAFER_1 |            |               |                |         |
| 5.1    |      | Enhanced B...  | pad nitride etch       | /Nitrides/Org...   |           | 40    |            |               | WAFER_1 | Top        | 0.05          |                |         |
| 5.2    |      | Enhanced B...  | pad oxide etch         | /Nitrides/Org...   |           | 4     |            |               | WAFER_1 | Top        | 0.05          |                |         |
| 5.3    |      | Enhanced B...  | silicon etch           | /Nitrides/Org...   |           | 100   |            |               | WAFER_1 | Top        | 0.05          |                |         |
| 6      |      | Remove M...    | resist strip           | /Organics/         |           |       |            |               | WAFER_1 |            |               |                |         |
| 7      |      | Conformal...   | sti fill deposition    | /Oxides/SiO2_H...  |           | 200   |            |               | WAFER_1 | Top        | 0.67          |                |         |
| 8      |      | Polish         | sti oxide cmp          |                    |           |       |            |               | WAFER_1 | Top        |               |                |         |

5. ASSIGNMENT1

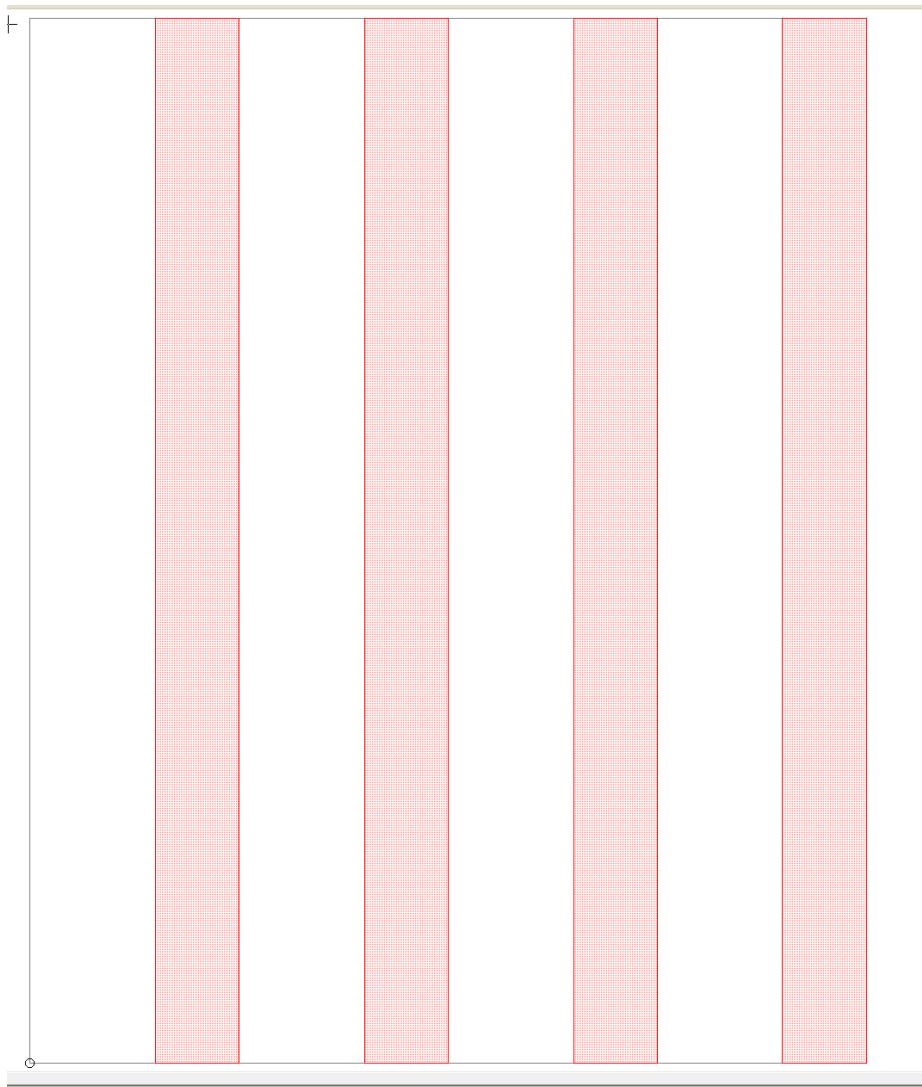


Figure 1: Assignment1 layout

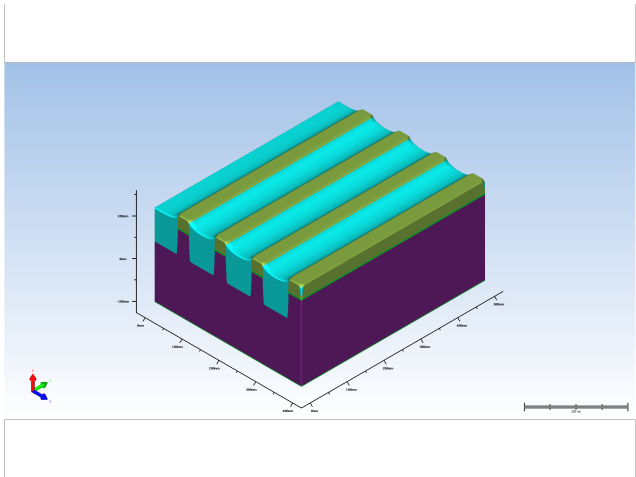


Figure 2: assignement 1 final output

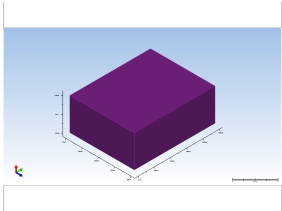


Figure 3

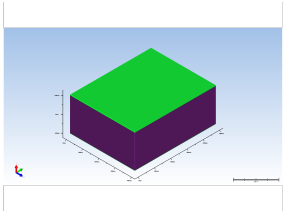


Figure 4

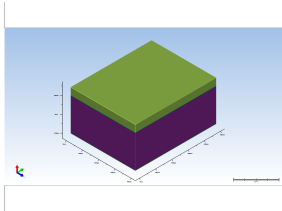


Figure 5

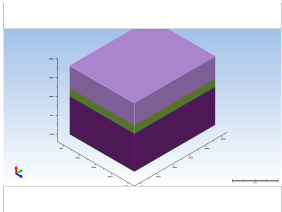


Figure 6

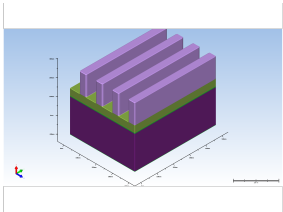


Figure 7

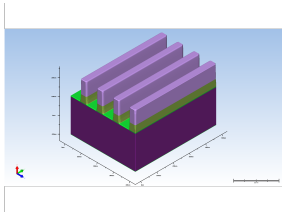


Figure 8

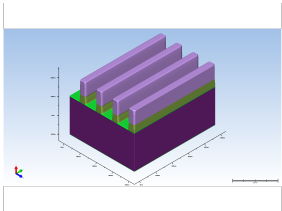


Figure 9

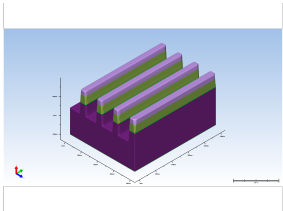


Figure 10

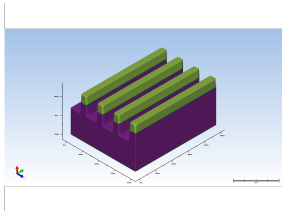


Figure 11

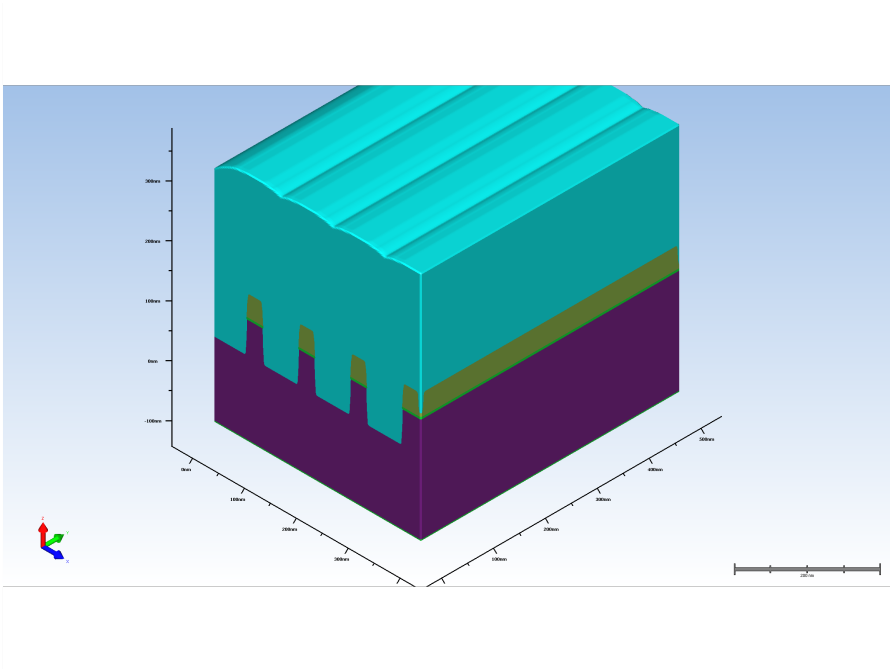


Figure 12

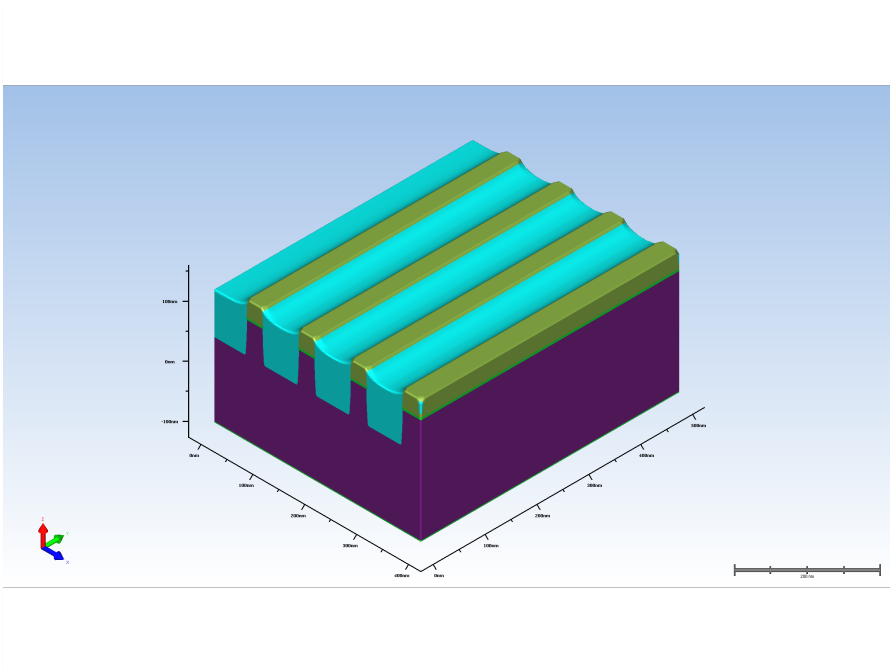


Figure 13: assignement 1 final output

6. ASSIGNMENT2

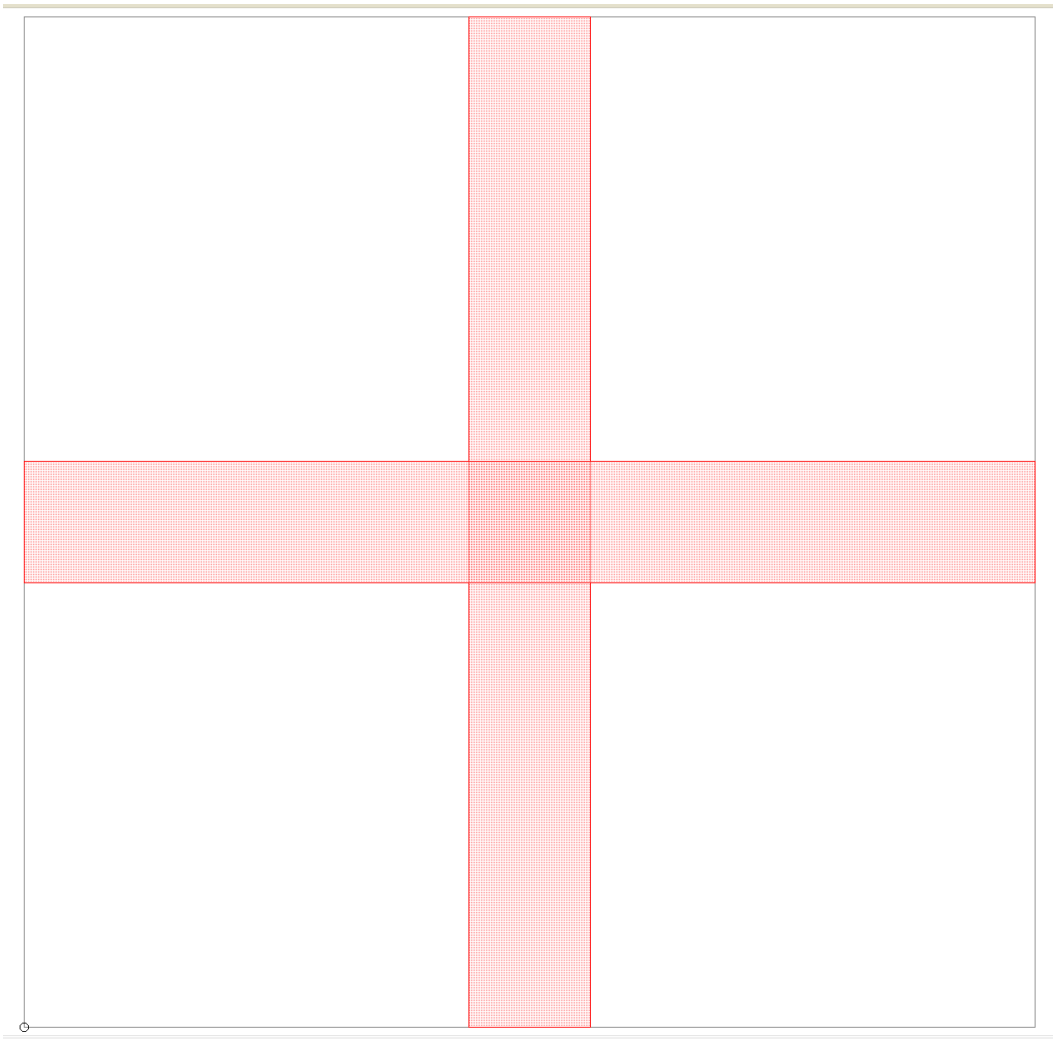


Figure 14: Assignment2 layout

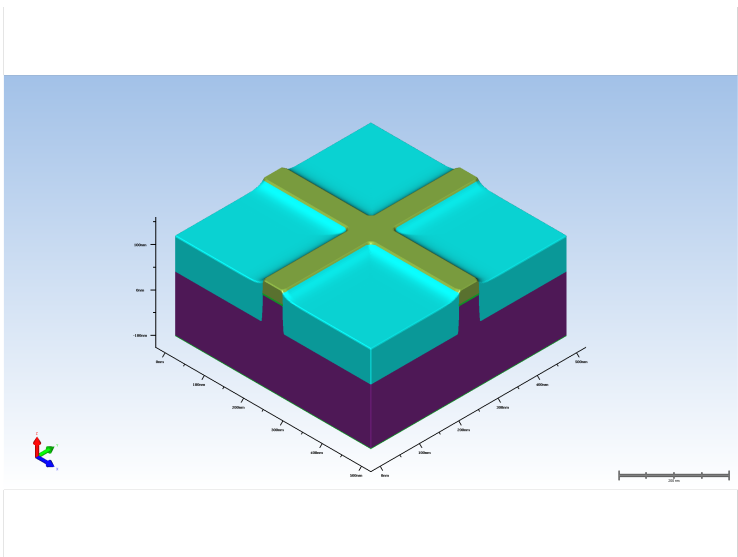


Figure 15: assignement 2 final output

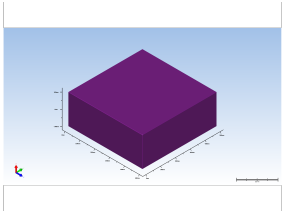


Figure 16

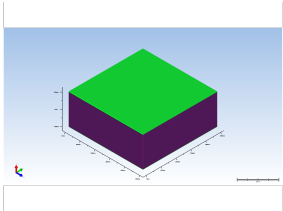


Figure 17

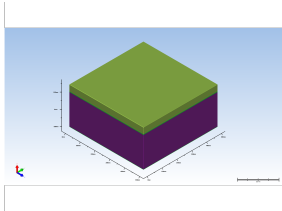


Figure 18

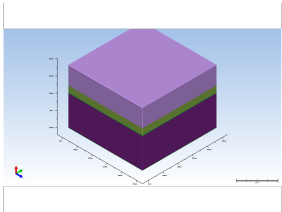


Figure 19

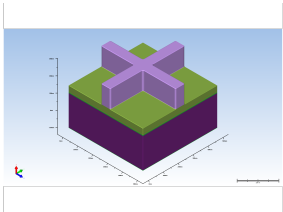


Figure 20

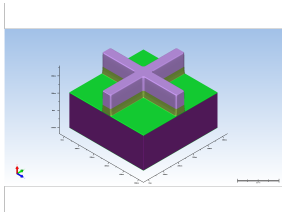


Figure 21

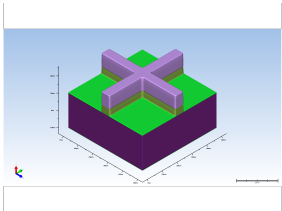


Figure 22

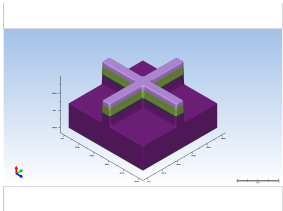


Figure 23

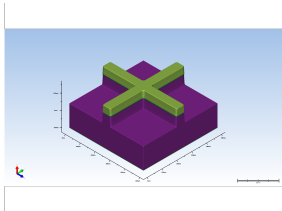


Figure 24

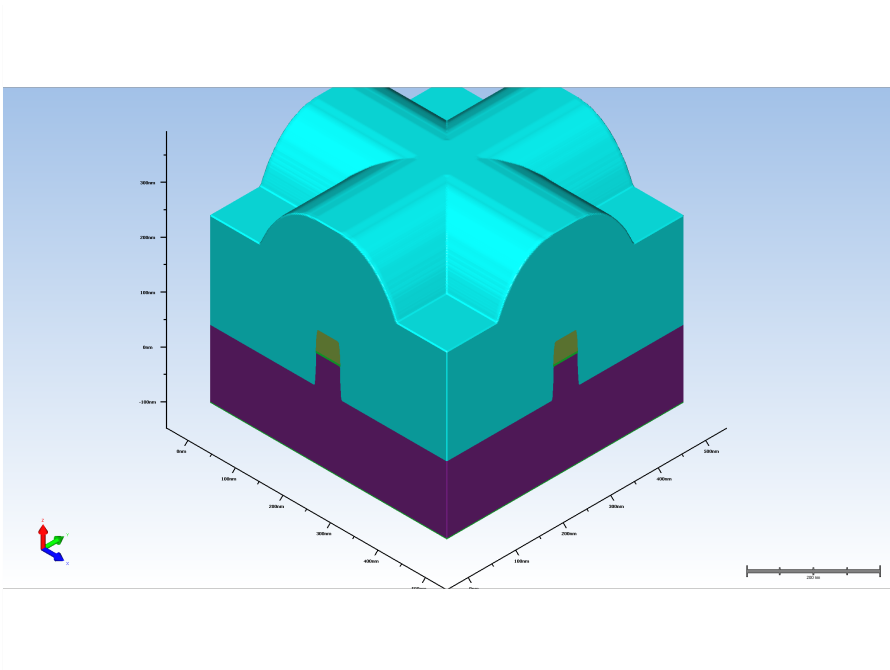


Figure 25

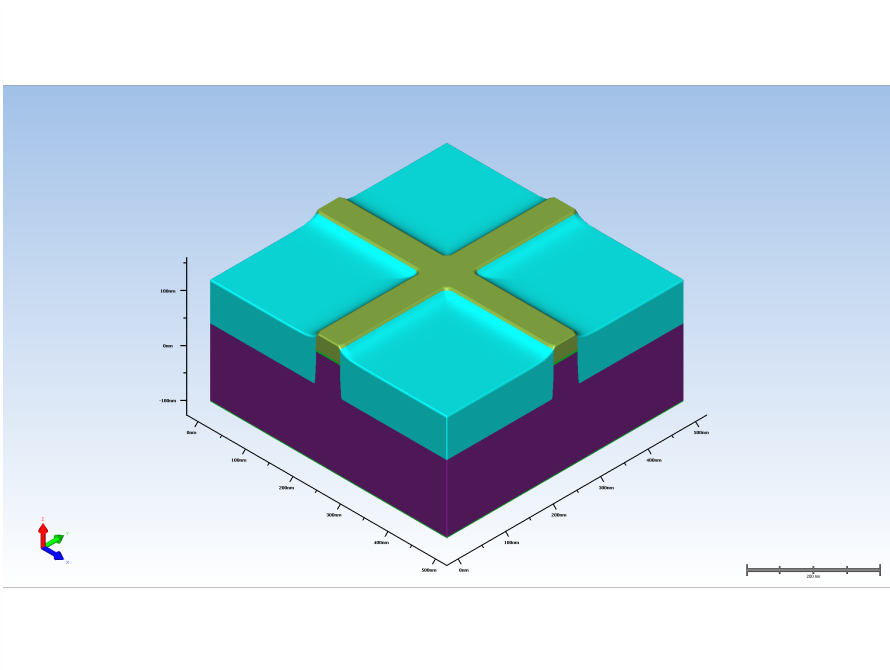


Figure 26: assignement 2 final output

## **7. OBSERVATION**

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Design 1 showed four uniform vertical line openings within the  $500\text{ nm} \times 400\text{ nm}$  boundary. Each line was  $40\text{ nm}$  wide with  $60\text{ nm}$  spacing, producing a regular line-space grating along the Y-axis.

Design 2 showed a cross-shaped pattern within the  $500\text{ nm} \times 500\text{ nm}$  boundary. The horizontal and vertical openings were each  $60\text{ nm}$  wide, extended across the full boundary, and intersected at the center.

## **8. CONCLUSION**

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The lithography mask patterns for shallow trench isolation were successfully designed and studied using Simulator3D. The parallel line array and orthogonal cross pattern were created with the specified dimensions, spacing, orientation, and boundary limits. The experiment demonstrated how accurately defined mask openings are used to represent isolation regions during semiconductor fabrication.